Mariem Kbaier

Final-Year Computer Engineering Student | AI & Data | Tunis, Tunisia

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PROFILE

Seeking a 2027 final-year internship (PFE) in AI & Data. Final-year ENIS student building LLM multi-agent systems, measurable RAG pipelines, and machine-learning models on real-world data, from analysis to reporting (Power BI) and deployment (Docker, CI/CD).

EXPERIENCE

AgriMent - Multi-Agent AI Agricultural Decision-Support Platform

Talan Tunisia - Summer Camp | Tunis | Jul-Aug 2026

LangGraph, Python, FastAPI, Mistral, Hybrid RAG (BM25 + RRF), Qdrant, ChromaDB, PyTorch, TempCNN, Sentinel-2, PostgreSQL/PostGIS, React, Docker

- Designed a 9-microservice decision-support platform for farmers and French buyers covering regulation, markets, and crop monitoring, with dynamic multi-agent routing, a supervisor, and agent-initiated handoffs. Placed 2nd among 10 teams (120+ participants).
- Built a regulatory hybrid-RAG pipeline (Mistral embeddings + BM25 + RRF) over Legifrance, FranceAgriMer, and CERFA forms. The best configuration reached 0.626 faithfulness (+14% vs. no RAG) and 0.941 context precision; expert evaluation validated RAGAS scores (Pearson r = 0.98, N = 6).
- Fine-tuned a TempCNN crop classifier on Sentinel-2 imagery (80.3% accuracy, kappa = 0.742, +1.8 pts vs. pretrained model) and designed a graduated-confidence architecture (LLM-grounded / ML-predicted / deterministic) with a numerical audit safeguard aligned with EU AI Act transparency rules.

Full-Stack Software Engineering Internship

BNS Engineering - Software Engineering Intern | Tunis | Jun-Aug 2025

Spring Boot, REST APIs, React.js, MySQL, Docker, Git, Agile/Scrum

- Developed a secure Spring Boot REST API (~15 endpoints) in a microservices architecture with a React.js frontend, MySQL, and Docker deployment for a self-service catalog (Agile/Scrum team).
- Integrated a semantic chatbot for natural-language catalog search and user support.

PROJECTS

- PerfPulse - Company Performance Analysis Agent (KPI & Power BI) (Academic project, ENIS)

Python, Pandas, LLMs, Power BI, DAX

- Designed an AI agent to analyze business KPIs, detect notable changes, identify the most affected product or region, and draft a concise natural-language conclusion.
- Separated deterministic computation (Python/Pandas) from LLM writing to keep figures reliable; presented and validated results in a Power BI dashboard.

- VORTEX - Multi-Agent BDD Test Automation (Final-year project, ENIS)

LangGraph, LangChain, Groq API, ChromaDB, Java, Spring Boot, Cucumber, RestAssured, JaCoCo, Maven

- Built a framework that turns business requirements into executable, self-healing test suites for Spring Boot microservices.
- Designed a 7-agent LangGraph pipeline combining RAG over 55,000+ BDD/JUnit scenarios with a self-repair loop: scenario coverage increased from 5% to 100%, and pass rate from 32% to 81.2%.
- Benchmarked three LLMs for output quality and token cost to select the best model for each agent.

- Agentic Job Matcher - CV-to-Job Matching System (Personal project - Talan Tunisia)

LangGraph, MCP (FastMCP), Apify API, Groq (Llama 3.3 70B), Sentence-Transformers, ChromaDB, Streamlit

- Built a candidate-to-job matching pipeline with evidence-backed explanations, supplied with live listings through the Apify API exposed as an MCP tool.
- Implemented deterministic scoring (embedding similarity + skills overlap), independent of LLM generation, and code-verified citations linking explanations to source text.
- Validated model changes with a golden dataset integrated into CI.

PROJECTS (CONTINUED)

- Company Logo Detection - Computer Vision (Personal project)
Python, YOLOv8, Ultralytics, PyTorch, Google Colab (GPU)
  - Fine-tuned YOLOv8s to detect 10 company logos across 379 annotated images: 0.995 mAP50, 0.98 precision, and 1.0 recall on validation, at ~4.5 ms/image inference (T4 GPU).

- Customer Behavior & Churn Prediction (Personal project)
Python, Pandas, Scikit-Learn, XGBoost, K-Means, DBSCAN, Random Forest, Flask
  - Built an end-to-end ML pipeline on 4,372 customers for churn prediction, segmentation, and spend forecasting.
  - Used XGBoost for churn (F1: 0.955), K-Means/DBSCAN for segmentation (silhouette: 0.822), and Random Forest for spend (R2 = 0.913).
  - Deployed the models through a Flask REST API with an interactive dashboard.

SKILLS

- AI & Machine Learning: LangGraph, LangChain, AI agents, RAG (hybrid search, BM25/RRF), multi-agent systems, MCP, LLM evaluation (RAGAS), Mistral, Groq, Sentence-Transformers, ChromaDB, Qdrant, PyTorch, scikit-learn, XGBoost, computer vision (YOLOv8)
- Data & BI: Pandas, SQL, Power BI, DAX, KPI analysis, PostgreSQL/PostGIS, MySQL, NoSQL
- MLOps & Cloud: Docker, Docker Compose, GitHub Actions (CI/CD), Git, AWS (Lambda, DynamoDB, IoT Core, S3)
- Languages & Frameworks: Python, Java, JavaScript, SQL; FastAPI, Flask, Spring Boot, React.js, Streamlit
- Concepts: Microservices, BDD/testing, MLOps, Agile/Scrum

EDUCATION

- ENIS - Engineering Degree in Computer Engineering | Sfax, Tunisia | 2024-Present
- IPEIT - Preparatory Cycle (Math, Physics, Computer Science) | Tunis, Tunisia | 2021-2024

AWARDS & ACTIVITIES

- 2nd place - Talan Summer Camp 2026: 2nd among 10 teams (120+ participants, ENIS) for AgriMent, a generative-AI agricultural decision-support platform.
- 1st place - Non-Technical Challenge: Represented IEEE PES Chapter - ENIS Student Branch as Webmaster.
- 1st place - TEC.Week 3.0: Tunisian Engineering Congress Week.

CERTIFICATIONS, ENGAGEMENT & LANGUAGES

- Certifications: LangGraph - Multi-Agent Systems & Building Scalable Agentic Systems (DataCamp) | Cisco C Advanced | Scrum Fundamentals (SCRUMstudy)
- Engagement: IEEE member - Pitch event organizer
- Languages: Arabic (native) | English (C1) | French (B2)